Tharaka A. Jayalath

tharaka.a.jayalath@gmail.com | +1-662-694-4520 | linkedin.com/in/tharaka-jayalath
github.com/taj771 | Google Scholar | taj771.github.io | Downingtown, PA

Professional Summary

Applied data scientist with a Ph.D. in Applied Economics, University of Saskatchewan. Core methods span demand modeling, causal inference, and predictive analytics on large behavioral and administrative datasets using Python, R, and SQL on Databricks.

Technical Skills

Languages	Python (pandas, scikit-learn, statsmodels), R (tidyverse, caret, lme4), SQL, MATLAB
Tools	Databricks, Jupyter Notebook, Tableau, Quarto, Pinecone, FastAPI, Streamlit, LaTeX
Methods	Causal Inference, Demand Modeling, Predictive Analytics, Discrete Choice / MNL, Experimental Design, Fixed Effects Regression, Monte Carlo Simulation, ETL Pipelines
Applied	Pricing Analytics, Demand Forecasting, Large-Scale Data Processing, Cross-functional Model Delivery, Stakeholder Communication, Generative AI

Education

Ph.D. in Applied Economics *2020 – 2025*
University of Saskatchewan, Saskatchewan, Canada

M.Sc. in Economics *2016 – 2018*
Mississippi State University, Mississippi, United States

B.Sc. in Agricultural Economics (Hons) *2008 – 2012*
Wayamba University, Sri Lanka

Certifications

The Data Science of Health Informatics · *Johns Hopkins University* 2026
Google Data Analytics · *Google* 2026
Agentic AI Builder's Bootcamp · *Maven* 2026
AI Engineering Bootcamp: Build AI Agents with Claude Code, RAG and MCP · *Maven* 2026
Capstone: full-stack species detection probability app — React/TypeScript frontend, Supabase/Postgres backend, RLS-secured schema, hourly forecast model. Live demo

Professional Experience

Research Data Scientist, University of Alabama *2026 – Present*
- Built ETL pipelines on Databricks in Python, transforming administrative databases into analysis-ready datasets for statistical modeling.
- Integrated USDA county-level data, commodity price series, and scraped retail indicators for multi-source economic program evaluation.
- Analyze multi-dimensional water, weather, and climate datasets using statistical and machine learning models; deliver quantitative outputs for federal agency program evaluation.
- Built USGS NWAA supply assessment for 3,609 U.S. data centers across 83,000+ HUC12 watersheds; outlet-routed HUC8 stress scores under greenfield and cumulative demand scenarios.
- Built Databricks pipeline computing 75 drought index configurations from ERA5 gridded data; Friedman tests quantified sensitivity of drought characterization to definition choice across 6 countries.

Research Data Scientist, University of Saskatchewan *2020 – 2025*
- Processed 3.5 million citizen-science records to estimate demand and recover price elasticity parameters from behavioral data in Python.
- Built individual-level RPL demand models on 3,850 observations, estimating heterogeneous preference distributions for program targeting.

Graduate Research Assistant, Mississippi State University *2016 – 2018*
- Joined U.S. Census covariates and USGS shapefiles to survey data (n=242) via ArcGIS spatial join for binary conservation enrollment classification.
- Fitted probit classification model to predict conservation program enrollment and estimate individual WTA compensation thresholds.

Management Trainee, Unilever Sri Lanka *2012 – 2013*
- Used SAP ERP to extract and structure financial data, producing budget variance and P&L performance reports in an enterprise finance setting.